YSP Clonidine Injection 150 mcg/mL

Ingredient(s):
Each ampoule (1 mL) contains:
Clonidine Hydrochloride ……………………………………………… 0.15 mg
Benzyl alcohol (as preservative) ………………………………………… 20.88 mg

Pharmacodynamics:
Mechanism of action
The hypotensive effect of Clonidine is produced mostly by its central effect or reducing sympathetic drive. In this respect Clonidine differs from previously used anti-hypertensives. Clonidine neither depletes major catecholamine stores, nor acts as a ganglion blocking agent. The specific and different mode of action of Clonidine leads to benefits such as reduced incidence of postural hypotension and only rarely an effect on libido.

The central action of Clonidine is ascribed mainly to an action on the bulbar structures of the central nervous system, particularly the sympathetic cardio-accelerator and constrictor mechanisms. This central action leads to decreased sympathetic outflow. Peripheral effects of Clonidine include both vasodilatation and vasoconstriction in various vascular beds, and alpha- and possible beta-adrenomimetic effects. A transient rise in blood sugar occurs following large doses of Clonidine. In addition, a small transient pressor effect (5-10 mm Hg systolic blood pressure) lasting approximately five minutes may occur following intravenous use. These effects reflect the alpha-adrenomimetic action of Clonidine. The peripheral effects of Clonidine generally require isolated organ type preparations for their demonstration, as in the intact animal or man, the central action predominates.

Clinical Trials
No data available

Pharmacokinetics:
Absorption and distribution
The pharmacokinetics of clonidine is dose-proportional in the range of 75-300 micrograms. Clonidine is well absorbed from the gastrointestinal tract and undergoes a minor first pass effect. Peak plasma concentrations are reached within 1-3 hours after oral administration. The duration of action varies from 6-12 hours, the duration of action being longer in the milder hypertensives. The plasma protein binding is 30-40%.

Metabolism and excretion
The terminal elimination half-life of clonidine has been found to range from 9-26 hours in patients with normal renal function. With impaired renal function it has been reported to increase to 18-48 hours.

The metabolic pathway of clonidine involves cleavage of the imidazolidine ring and the hydroxylation of the phenyl ring. Five metabolites have been identified in man and include para-hydroxy-clonidine and dichlorophenylguanidine.

Two thirds of an administered dose is excreted in the urine (about half of which is unchanged Clonidine) and the remainder is excreted in the faeces.

The antihypertensive effect is reached at plasma concentrations between about 0.2 and 2.0 ng/mL in patients with normal renal function. The hypotensive effect is attenuated or decreases with plasma concentrations above 2.0 ng/mL.

Given intravenously Clonidine is effective within five minutes, has a maximum hypotensive action within 20 to 30 minutes, and the effect lasts for several hours. Following intramuscular administration, Clonidine is effective within 5 to 10 minutes. The maximum hypotensive effect is reached after 75 minutes, and the duration of action is approximately 5 hours.

Indication(s):
Acute hypertensive crisis. As an alternative to oral therapy where the oral route of administration is inappropriate.

Dosage and Administration(s):
Intramuscular injection should only be administered to patients in a lying position.

One to two ampoules (150-300 mcg) by intramuscular injection undiluted, or given intravenously with 10 mL of normal saline over 5 minutes. May be repeated at intervals of 3 to 6 hours as necessary. Following intravenous injection, an initial pressor phase of 5-10 mm Hg lasting approximately 5 minutes may occur. This effect can be lessened by slow administration. As clonidine is metabolised in the liver and excreted mainly by the kidneys, any hepatic or renal impairment may require a reduction in dosage. This product may be given in combination with guanethidine, alpha methyldopa or other antihypertensives to provide effective control of blood pressure in refractory cases. In this way the dose of each individual drug may be reduced and side effects minimised.

Compatible diluent	Final concentration after dilution	
Normal saline solution (0.9% NaCl)	Minimum concentration	1 amp. with 10 mL of normal saline means that 0.15 mg Clonidine Hydrochloride with 10 mL of normal saline, equivalent to 0.015 mg/mL of Clonidine Hydrochloride in normal saline solution.
	Maximum concentration	2 amp. with 10 mL of normal saline means that 0.30 mg Clonidine Hydrochloride with 10 mL of normal saline, equivalent to 0.030 mg/mL of Clonidine Hydrochloride in normal saline solution.

Renal impairment

Dosage must be adjusted:

- according to the individual antihypertensive response which can show high variability in patients with renal insufficiency.
- according to the degree of renal impairment.

Careful monitoring is required. Since only a minimal amount of clonidine is removed during routine haemodialysis, there is no need to give supplemental clonidine following dialysis.

Route of Administration(s):

For intramuscular (IM) or intravenous (IV) use.

Contraindication:

Clonidine should not be used in patients with known hypersensitivity to the active ingredient, clonidine hydrochloride, and in patients with severe bradyarrhythmia resulting from either sick sinus syndrome or AV block of second or third degree.

In case of rare hereditary conditions that may be incompatible with an excipient of the product the use of the product is contraindicated.

Warning(s) and Precaution(s):

Special care should be exercised in treating patients who have a history of depression or who have advanced cerebrovascular disease. Reduction of blood pressure in the latter circumstances may itself cause mental changes.

Concurrent administration of tricyclic antidepressants may require adjustment of Clonidine dosage.

Although a transient rise in blood sugar has been noted occasionally in humans treated with Clonidine, which may be due to a pharmacologic alpha-adrenomimetic effect of the drug, no case of induced diabetes mellitus due to Clonidine has been reported. Patients with clinical diabetes mellitus should be watched for a possible increase in their requirements of anti-diabetic therapy.

Clonidine should be used with caution in patients with mild to moderate bradyarrhythmia such as low sinus rhythm, with disorders of cerebral or peripheral perfusion, polyneuropathy, and constipation.

No therapeutic effect of Clonidine can be expected in the treatment of hypertension caused by phaeochromocytoma.

Since Clonidine, and its metabolites are extensively excreted in the urine, careful adjustment of dosage is required in patients with renal insufficiency.

As with other anti-hypertensives, treatment with Clonidine should be monitored particularly carefully in patients with heart failure or severe coronary heart disease.

Termination of oral therapy should be gradual (e.g. over more than 7 days).

Sudden cessation of antihypertensive therapy is known to be associated in some instances with rebound hypertension which in some cases may be severe. This may occur with Clonidine particularly in patients receiving more than the maximum recommended dose of 900 micrograms per day.

Following sudden discontinuation of Clonidine after prolonged treatment with high doses, restlessness, palpitations, rapid rise in blood pressure, nervousness, tremor, headache, or nausea have been reported.

An excessive rise in blood pressure following discontinuation of Clonidine

therapy can be reversed by intravenous phentolamine.
If long-term treatment with a β-blocker needs to be interrupted, the β-blocker should be gradually phased out first, then clonidine.
Patients who wear contact lenses should be warned that treatment with Clonidine may cause decreased lacrimation.
Patients with the rare hereditary conditions of galactose intolerance e.g. galactosaemia should not take this medicine.

Anaesthesia
Abrupt withdrawal of Clonidine is undesirable. Limited evidence suggests that it is unnecessary to withdraw Clonidine before anaesthesia and that maintenance of therapy is preferable to abrupt withdrawal. In the peri-operative period Clonidine can, where necessary, be administered parenterally until oral therapy is resumed.
Where therapy with Clonidine is to be suspended before operation, withdrawal should be gradual (i.e. over more than 7 days) and monitored by regular observation of blood pressure.

Use in the elderly
No data available.

Paediatric use
The use and the safety of clonidine in children and adolescents has little supporting evidence in randomised controlled trials and therefore cannot be recommended for use in this population.
In particular, when clonidine is used off-label concomitantly with methylphenidate in children with ADHD, serious adverse reactions, including death, have been observed. Therefore, clonidine in this combination is not recommended.

Effects on laboratory tests
No data available.

Warning:
As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

Interaction with Other Medicament(s):
If the patient is on antihypertensive therapy, care should be taken as even a small dose of clonidine may further lower blood pressure and necessitate adjustment of the antihypertensive regime.

When Clonidine is used as an antihypertensive agent additional clonidine for the prophylaxis of migraine or the alleviation of symptoms in menopausal flushing should not be prescribed. Clonidine may potentiate the effects of alcohol, sedatives, hypnotics or other centrally active substances.
Although retinal, lens or corneal damage have not been detected with clonidine therapy, follow up procedures, such as ophthalmoscopy, are recommended.

Substances which raise blood pressure or induce a sodium and water retaining effect such as nonsteroidal anti-inflammatory drugs can reduce the therapeutic effect of clonidine.

Substances with α2-adrenergic receptor blocking properties, such as phentolamine, may abolish the α2-adrenergic receptor mediated effects of clonidine in a dose-dependent way.
Concomitant administration of drugs with a negative chronotropic or dromotropic effect such as β-blockers or digitalis glycosides can cause or potentiate bradycardic rhythm disturbances.

It cannot be ruled out that concomitant administration of a β-blocker will cause or potentiate peripheral vascular disorders.

The antihypertensive effect of clonidine may be reduced or abolished, and orthostatic regulation disturbances may be provoked or aggravated by concomitant administration of tricyclic antidepressants or neuroleptics with α-receptor blocking effects.
Based on observations in patients in a state of delirium alcoholicum, it has been suggested that high intravenous doses of clonidine may increase the arrhythmogenic potential (QT-prolongation, ventricular. fibrillation) of high intravenous doses of haloperidol.

Pregnancy:
Clonidine hydrochloride has not shown teratogenic potential when tested in

rats, but in some circumstances the incidence of embryonic and perinatal deaths was increased with doses comparable to those used clinically for antihypertensive therapy.
There are limited data from the use of clonidine in pregnant women, but the experience with clonidine hydrochloride since marketing does not include any positive evidence of adverse effect on the foetus. Since this experience cannot exclude such an effect, clonidine hydrochloride should be used during pregnancy only when the benefit clearly justifies the possible risk to the foetus.
Clonidine passes the placental barrier and may lower the heart rate of the foetus. There is no adequate experience regarding the long-term effects of prenatal exposure.
Clonidine hydrochloride may also induce transitory elevation of blood glucose and impairment of glucose tolerance. Children born to mothers treated with clonidine hydrochloride during pregnancy should be specifically examined for changes in glucose metabolism.
During pregnancy the oral forms of clonidine are preferred. Intravenous injection of clonidine should be avoided.
Non-clinical studies in rats do not indicate direct or indirect harmful effects with respect to reproductive toxicity.
Postpartum a transient rise in blood pressure in the newborn cannot be excluded.

Fertility:
Clinical studies on the effect of clonidine on human fertility have not been conducted.

Clonidine had no effect on fertility in male or female rats when administered orally at doses up to 0.15 mg/kg/day (35% higher than the maximum recommended total daily dose of clonidine in humans, based on body surface area).

Lactation:
Clonidine is excreted in human milk. As the effect on the newborn is not known, infants born to mothers being treated with Clonidine should not be breast fed.

Side effects:
The corresponding frequency category estimation for each adverse drug reaction is based on the following convention: very common (≥1/10); common (≥1/100 to <1/10); uncommon (≥1/1,000 to <1/100); rare (≥ 1/10,000 to <1/1,000); very rare (<1/10,000); not known (cannot be estimated from the available data).

System Organ Class	Very Common	Common	Uncommon	Rare	Not Known
Endocrine disorders:				gynaecomastia	
Psychiatric disorders:		depression, sleep disorder	delusional perception, hallucination, nightmare		confusional state, libido decreased
Nervous system disorders:	dizziness, sedation	headache	paraesthesia		
Eye disorder:				lacrimation decreased	accommodation disorder
Cardiac disorders:			sinus bradycardia	atrioventricular block	bradyarrhythmia
Vascular disorders:	orthostatic hypotension		Raynaud's phenomenon		
Respiratory, thoracic and mediastinal disorders:				nasal dryness	
Gastrointestinal disorders:	dry mouth	constipation, nausea, salivary gland pain, vomiting		colonic pseudo -obstruction	
Skin and subcutaneous tissue disorders:			pruritus, rash, urticaria	alopecia	

Reproductive system and breast disorders:		erectile dysfunction			
General disorders and administration site conditions:		fatigue	malaise		
Investigations:				blood glucose increased	

Most adverse effects are mild and tend to diminish with continued therapy. Occasional reports of abnormal liver function tests and cases of hepatitis have also been reported.

Effects on Ability to Drive and Use Machine:
No studies on the effects on the ability to drive and use machines have been performed.

However, patients should be advised that they may experience undesirable effects such as dizziness, sedation, and accommodation disorder during treatment with Clonidine. Therefore, caution should be recommended when driving a car or operating machinery. If patients experience the above mentioned side effects, they should avoid potentially hazardous tasks such as driving or operating machinery.

Symptoms and Treatment of Overdose:
Symptoms
The most important features of clonidine overdosage are likely to be bradycardia, sedation, respiratory depression including apnoea and somnolence including coma. Blood pressure response may be variable and may vary from severe hypotension (due to central sympathetic inhibition and vagal stimulation) to severe hypertension (due to direct alpha agonist activity). Treatment must therefore be appropriate to the clinical features (i.v. atropine followed by a pressor amine if necessary, in patients with hypotension or an alpha blocker such as phentolamine for patients with hypertension). Other features which may be seen include weakness, vomiting, diminished or absent reflexes, skin pallor, hypothermia, cardiac arryhthmias and constricted pupils with poor reaction to light.

Management
General supportive measures with regular checks of pulse, B.P., ECG, blood sugar and body temperature should be undertaken. The blood pressure should be monitored carefully for 48 hours following the overdosage, as a later hypertensive phase may be associated with declining blood levels of clonidine.

Incompatibilities:
Incompatibilities were either not assessed or not identified as part of the registration of this medicine.

Shelf Life:
Unopened ampoule: 3 years from the date of manufacture.
After dilution for infusion administration: Store at temperature below 30°C for up to 48 hours.

Storage Condition:
Unopened ampoule: Store at temperature below 30°C.
After dilution for infusion administration: Store at temperature below 30°C for up to 48 hours.

Product Description:
A clear and colourless solution.

Dosage Forms and Packaging Available:
1 mL x 5 ampoules

Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
Lot 3, 5, 7 & 12, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
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Ordering Line: 1 800 88 3027
Product Info: 1 800 88 3679

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